

DCF Valuation of BMW - Practice 7

Based on the restrictions imposed in the Practice instructions (see below), Yahoo Finance was used to screen for a list of companies out of which *Bayerische Motoren Werke AG* was selected.

- Screen initially for:
 - **Price to Earnings (P/E)** ratio no greater than 20
 - The Company's **Return on Equity (ROE)** should be greater than or equal to 10%
- Secondary Screen
 - Price to Book less than 1.5 – 2.0
 - Debt to Equity Ratio less than 1.5 – 2.0
 - Current Ratio greater or equal to 1.0

Imposed restrictions for company selection

BMW's Balance Sheet, Income Statement, and relevant Cash Flow Statement line items (Depreciation & amortization, CAPEX, and Increases in Working Capital) were downloaded for the last 5 years (2019-2023).

BMW Income Statement (in € million)	2019	2020	2021	2022	2023
Revenues	104.210	98.990	111.239	142.610	155.498
Cost of sales	(86.147)	(85.408)	(89.253)	(118.042)	(125.809)
Gross profit	18.063	13.582	21.986	24.568	29.689
Selling and administrative expenses	(9.367)	(8.795)	(9.233)	(10.616)	(11.025)
Other operating income and expenses	(1.285)	43	647	47	(182)
Profit before financial result	7.411	4.830	13.400	13.999	18.482
Financial result	(293)	392	2.660	9.510	(1.386)
Profit before tax	7.118	5.222	16.060	23.509	17.096
Income taxes	(2.140)	(1.365)	(3.597)	(4.927)	(4.931)
Net profit	4.978	3.857	12.463	18.582	12.165

BMW Income Statement

BMW Balance Sheet (in € million)	2019	2020	2021	2022	2023
Intangible assets	11.729	12.342	12.980	21.776	20.022
Property, plant and equipment	23.245	21.850	22.390	32.126	35.266
Leased products	42.609	41.995	44.700	42.820	43.118
Investments accounted for using the equity method	3.199	3.585	5.112	420	443
Other investments	703	735	1.241	1.351	1.197
Receivables from sales financing	92.437	84.277	87.417	85.708	87.355
Financial assets	7.325	7.752	7.515	8.237	5.518
Deferred and current tax	3.403	3.065	3.731	2.854	3.630
Other assets	12.939	10.326	10.243	10.632	9.133
Inventories	15.891	14.896	15.928	20.005	23.719
Trade receivables	2.518	2.298	2.261	4.127	4.162
Cash and cash equivalents	12.036	13.537	16.009	16.870	17.327
Total assets	228.034	216.658	229.527	246.926	250.890
Equity	59.907	61.520	75.132	91.288	92.923
Pension provisions	3.335	3.693	1.247	339	427
Other provisions	13.209	13.982	13.954	15.761	17.037
Deferred and current tax	1.595	1.256	2.379	3.989	4.198
Financial liabilities	116.740	106.376	103.463	94.196	95.010
Trade payables	10.182	8.644	10.932	14.120	15.547
Other liabilities	23.066	21.187	22.420	27.233	25.748
Total equity and liabilities	228.034	216.658	229.527	246.926	250.890

BMW Balance Sheet

CAPEX	(6.902)	(6.150)	(6.619)	(9.050)	(10.881)
Depreciation and amortisation of tangible and intangible assets	6.017	6.139	6.495	8.566	8.974
Changes in Working Capital	(1.117)	(996)	1.355	(523)	(2.715)

BMW Cash Flow Statement relevant items

OUR ASSUMPTIONS

The following assumptions were made based on these financial statements:

Assumptions:						Average or 2024	2025	2026	2027	2028
Growth rate of revenues	7,6%	-5,0%	12,4%	28,2%	9,0%	10,4%	8,7%	6,9%	5,2%	3,4%
Cost of sales/Revenues	-82,7%	-86,3%	-80,2%	-82,8%	-80,9%	-81,6%	-81,6%	-81,6%	-81,6%	-81,6%
Selling and administrative expenses/Revenues	-9,0%	-8,9%	-8,3%	-7,4%	-7,1%	-7,0%	-7,0%	-6,9%	-6,9%	-6,8%
Other operating income and expenses/Revenues	-1,23%	0,04%	0,58%	0,03%	-0,12%	-0,14%	-0,14%	-0,14%	-0,14%	-0,14%
EBIT Margin	7,1%	4,9%	12,0%	9,8%	11,9%	11,2%	11,2%	11,3%	11,3%	11,4%
PPE+Intangible Assets	34.974	34.192	35.370	53.902	55.288	57.422	60.229	63.419	66.827	70.225
CAPEX/Revenues	6,6%	6,2%	6,0%	6,3%	7,0%	6,4%	6,4%	6,4%	6,4%	6,4%
Depreciation and amortisation of tangible and intangible assets/Fixed Assets(t-1)	-19,6%	-18,0%	-18,4%	-15,9%	-16,2%	-16,10%	-16,00%	-16,00%	-15,90%	-15,80%
Changes in Working Capital/Revenues	-1,07%	-1,01%	1,22%	-0,37%	-1,75%	-0,59%	-0,59%	-0,59%	-0,59%	-0,59%
Corporate tax rate in Germany						30%	30%	30%	30%	30%
Net Profit Margin	4,78%	3,90%	11,20%	13,03%	7,82%	8,15%	8,82%	9,80%	9,52%	8,82%

Assumptions made to obtain FCFs

1. The growth rate of revenues in 2024 was assumed to be the average growth rate of the last five years. From then on, based on EU regulations imposing EVs, and increasing competition from other traditional car manufacturers and Chinese EVs penetrating the European market, revenue growth was assumed to decrease by 1,75% each year until 2028.
2. Cost of sales as a percentage of revenue was assumed to remain constant and equal to the average of the last five years in the projected years.
3. Selling and Administrative expenses as a function of revenue were assumed to keep decreasing in the trend they were over the last five years.
4. Other Operating income and expenses as a percentage of revenue were assumed to be constant and equal to the average of the last five years in the projected period.
5. EBIT Margins for the projected years were obtained by subtracting the previous expenses as a percentage of revenue from 1 for each projected year.
6. CAPEX as a percentage of revenues was assumed to be constant and equal to the average of the last five years in the projected period.
7. Depreciation and amortization was linked to PPE+Intangible Assets from the prior year, and a decreasing trend similar to the one over the last five years was assumed.
8. A schedule was built to forecast PPE+Intangible Assets for each of the projected years by adding the current year's CAPEX and subtracting the current year's Depreciation and Amortization from the PPE+Intangible Assets at the end of the prior year.
9. Changes in Working Capital over Revenues were assumed to be constant and equal to their average over the last five years for the projected period.
10. A constant corporate tax rate of 30% was assumed for the projected period.

Forecasting Free Cash Flows (FCF)

Based on these assumptions, Free Cash Flows to the firm were projected.

<i>Free Cash Flows</i> (in € million)	2024	2025	2026	2027	2028
EBIT(1- T_c)	13.483	14.655	15.811	16.632	17.356
Depreciation	8.901	9.188	9.637	10.084	10.559
CAPEX	(11.035)	(11.994)	(12.827)	(13.492)	(13.956)
Increases in NWC	1.021	1.110	1.187	1.248	1.291
FCF	12.370	12.958	13.808	14.472	15.250

Projected Free Cash Flows to the Firm

- 2024: €13,483 million
- 2025: €14,655 million
- 2026: €15,811 million
- 2027: €16,632 million
- 2028: €17,356 million

Discount Rate - Weighted Average Cost of Capital (WACC)

- The WACC serves as the discount rate for future cash flows and is calculated as a weighted average of the cost of debt and the cost of equity, reflecting the company's capital structure:
 - **Cost of Equity (Calculated using CAPM):** The Capital Asset Pricing Model

$$R_i = R_f + \beta_i * (R_m - R_f)$$

(CAPM) is used here, with the following parameters:

- **Risk-Free Rate:** 2.815%, based on the current German Government Schatz-Bund 3 months yield rate.

- **Stock Beta:** 1.08, representing BMW's volatility relative to the broader market monthly in the last 5 years. (yahoo finance)
- **Market Risk Premium:** using the German market return, following DAX40 Index, we assumed an expected return equal to YTD's one (23.1968%), which, when combined with the risk-free rate and beta, gives a **Cost of Equity** of 27.87%.
- **Cost of Debt:** 4.235%, derived from the risk-free rate plus a spread of 1.42%, what we did was adding to the current risk free rate the credit default spread corresponding to the Moody's credit rate of A2 on BMW, which resulted in 1.42%.
- **Capital Structure Weights:**
 - **Debt:** 62.96% of total assets.
 - **Equity:** 37.04% of total assets.

Using these inputs, the model calculates the **WACC at 12.19%**.

WACC Calculation:

risk free rate	2,82%
spread	1,42%
Cost of debt	4,24%
Liabilities/Total Assets 2023	63%

stock Beta	1,08
Market risk premium	23,20%
Cost of Equity	27,87%
Equity/Total Assets 2023	37%

WACC	12,19%
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g	1%
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$$WACC = w_d r_d (1-t) + w_e r_e$$

Diagram illustrating the WACC formula components:

- w_d : Weight of Debt
- r_d : Cost of Debt
- $(1-t)$: Effective Tax Rate
- w_e : Weight of Equity
- r_e : Cost of Equity

Discount Factor and Present Value of FCFs

- Each year's FCF is discounted using the WACC (12.19%) as the discount rate, resulting in the following discount factors:
 - 2024: 0.891
 - 2025: 0.795
 - 2026: 0.708
 - 2027: 0.631
 - 2028: 0.563

- **Present Value of FCFs:** Each FCF is multiplied by its respective discount factor, adjusting each future cash flow to its present value.

Calculating the Terminal Value (TV)

- **Terminal Growth Rate (g):** Set at 1%, reflecting conservative growth beyond the forecast period (taking into account that the automotive industry is a mature industry).

- **Terminal Value Calculation:**

$$TV = (FCF_{2028} * (1 + g) / WACC - g) / (0.1219 - 0.01) \approx 156.683 \text{ millions}$$

- **Discounted Terminal Value:** Applying the 2028 discount factor of 0.563 results in the terminal value's present value, which contributes to the overall enterprise value.

Calculating the Enterprise Value (EV) and Stock Price

- **Enterprise Value (EV):** Adding the present values of all FCFs from 2024 to 2028 plus the discounted terminal value gives an **Enterprise Value of €143,289 million**.

Adjusting the Enterprise Value by subtracting Net Financial Debt and Provisions, the Equity Value was obtained.

- **Equity Value and Stock Price:**

Finally, the Equity Value was divided by Weighted Average Ordinary Shares Outstanding (from BMW's last quarterly report) to obtain the Fair Value of BMW's Share Price.

Margin of Safety

With the resulting fair value of BMW's stock, we establish a margin of safety by creating an interval 20% below the estimated intrinsic value. This margin helps account for potential uncertainties in the valuation. The current BMW price is approximately 5% different from the final calculated value. The fair value interval, incorporating the 20% margin of safety, ranges from **€61 to €91**.

DCF	2024	2025	2026	2027	2028	TV 2028
FCF	13.483	14.655	15.811	16.632	17.356	156.683
discount factor	0,891	0,795	0,708	0,631	0,563	0,563

Enterprise Value	143.289
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Net Financial Debt	77.683
Provisions	17.464

WACC	12,19%
g	1%

Equity Value	48.142
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Shares outstanding	630.000.000
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Stock price (Fair Value)	76,42 €
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Current stock price	73,22 €
Current Difference (%)	4,18%
MARGIN OF SAFETY	30%
Acceptable buy price (lower limit)	53,49 €
Upper limit	99,34 €

Conclusion

1. **Investment Opportunity:** The stock appears to be trading below its intrinsic value, which may indicate a buying opportunity for investors seeking to purchase at a discount. If the fundamentals align with this valuation, BMW might be expected to appreciate its fair value over time.
2. **Margin of Safety:** Given that the FV is higher, the margin of safety is likely met or exceeded, adding a buffer for potential risks or uncertainties in the assumptions. This strengthens the case for investment, as the downside risk is somewhat mitigated.
3. **Market Sentiment or Short-Term Factors:** The lower current price could be due to temporary market factors, sentiment shifts, or macroeconomic uncertainties affecting BMW's stock.
If these factors are transient and unrelated to the company's long-term value, they may present a favorable entry point.
4. **Further Due Diligence:** Before acting, it's advisable to review assumptions like growth rates, WACC, and market conditions. Ensuring these are conservative and realistic will confirm whether the undervaluation reflects a genuine opportunity or if adjustments in assumptions are necessary.